T3 C_+1: general EFT1 RGEs

Matchete/RGBeta symbolic output in mathematical notation

C_+1

Additional quartics: 1, additional bilinears: 0, extra Yukawa: none.
General matching and Wilson coefficients are not computed in this symbolic report.

One-loop renormalisable beta functions

$16\pi^2\, dX/d\ln\mu = \beta_X$

g_1

$$16\pi^2 \frac{dg_1}{d\ln\mu} = \frac{17{g_1}^3}{2}$$

g_2

$$16\pi^2 \frac{dg_2}{d\ln\mu} = -\frac{17{g_2}^3}{6}$$

g_3

$$16\pi^2 \frac{dg_3}{d\ln\mu} = -7{g_3}^3$$

Y_u

$$\begin{aligned}
16\pi^2 \frac{dY_u}{d\ln\mu} = {} & 3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_u)_{ij} \\
& + \mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_u)_{ij} \\
& + 3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_u)_{ij} \\
& - \frac{3}{2}(\Big(Y_d\ (Y_d)^\dagger\ Y_u\Big)_{ij} - \Big(Y_u\ (Y_u)^\dagger\ Y_u\Big)_{ij}) \\
& - \frac{9}{4}{g_2}^2\,(Y_u)_{ij} \\
& - 8{g_3}^2\,(Y_u)_{ij} \\
& - \frac{17}{12}{g_1}^2\,(Y_u)_{ij}
\end{aligned}$$

Y_d

$$\begin{aligned}
16\pi^2 \frac{dY_d}{d\ln\mu} = {} & \mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij} \\
& + 3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij} \\
& + 3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij} \\
& + \frac{3}{2}(\Big(Y_d\ (Y_d)^\dagger\ Y_d\Big)_{ij} - \Big(Y_u\ (Y_u)^\dagger\ Y_d\Big)_{ij}) \\
& - \frac{9}{4}{g_2}^2\,(Y_d)_{ij} \\
& - 8{g_3}^2\,(Y_d)_{ij} \\
& - \frac{5}{12}{g_1}^2\,(Y_d)_{ij}
\end{aligned}$$

Y_e

$$\begin{aligned}16\pi^2 \frac{dY_e}{d\ln\mu} = & 3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_e)_{ij} \\ & + 3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_e)_{ij} \\ & + \mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_e)_{ij} \\ & + \frac{3}{2}\Big(Y_e\ (Y_e)^\dagger\ Y_e\Big)_{ij} \\ & - \frac{9}{4}{g_2}^2\,(Y_e)_{ij} \\ & - \frac{15}{4}{g_1}^2\,(Y_e)_{ij}\end{aligned}$$

m_1^2

$$\begin{aligned}16\pi^2 \frac{dm_1^2}{d\ln\mu} = & -\frac{9{g_2}^2 m_1^2}{2} \\ & - \frac{3{g_1}^2 m_1^2}{2} \\ & + 4\lambda_{12}^{(1)} m_2^2 \\ & + 6\lambda_{S_1}^{(1)} m_1^2\end{aligned}$$

m_2^2

$$\begin{aligned}16\pi^2 \frac{dm_2^2}{d\ln\mu} = & -\frac{9{g_2}^2 m_2^2}{2} \\ & - \frac{27{g_1}^2 m_2^2}{2} \\ & + 4\lambda_{12}^{(1)} m_1^2 \\ & + 6\lambda_{S_2}^{(1)} m_2^2\end{aligned}$$

λ_1

$$\begin{aligned}16\pi^2 \frac{d\lambda_1}{d\ln\mu} = & \lambda_1(12\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 4\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 12\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)) \\ & - 12\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\Big) \\ & - 4\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\Big) \\ & - 12\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\Big) \\ & + 4\lambda_5\,(\lambda_5)^* \\ & + 8\mathrm{lambdaT3GeneralQ022000I1}\,(\mathrm{lambdaT3GeneralQ022000I1})^* \\ & + \frac{9{g_2}^4}{4} \\ & + {g_1}^2\left(\frac{3{g_2}^2}{2} - 3\lambda_1\right) \\ & - 9{g_2}^2\lambda_1 \\ & + \frac{3{g_1}^4}{4}\end{aligned}$$

$$
\begin{aligned}
&+ 12{\lambda_1}^2 \\
&+ 4{\lambda_{HS_1}^{(1)}}^2 \\
&+ \frac{{\lambda_{HS_1}^{(A)}}^2}{4} \\
&+ 4{\lambda_{HS_2}^{(1)}}^2 \\
&+ \frac{{\lambda_{HS_2}^{(A)}}^2}{4}
\end{aligned}
$$

$\lambda_{S_1}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{S_1}^{(1)}}{d\ln\mu} = {} & 8\text{lambdaT3GeneralQ022000I1 } (\text{lambdaT3GeneralQ022000I1})^* \\
&+ \frac{9{g_2}^4}{4} \\
&+ {g_1}^2 \left(\frac{3{g_2}^2}{2} - 3\lambda_{S_1}^{(1)} \right) \\
&- 9{g_2}^2 \lambda_{S_1}^{(1)} \\
&+ \frac{3{g_1}^4}{4} \\
&+ 4{\lambda_{12}^{(1)}}^2 \\
&+ \frac{{\lambda_{12}^{(A)}}^2}{4} \\
&+ 4{\lambda_{HS_1}^{(1)}}^2 \\
&+ \frac{{\lambda_{HS_1}^{(A)}}^2}{4} \\
&+ 12{\lambda_{S_1}^{(1)}}^2
\end{aligned}
$$

$\lambda_{S_2}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{S_2}^{(1)}}{d\ln\mu} = {} & \frac{9{g_2}^4}{4} \\
&+ {g_1}^2 \left(\frac{27{g_2}^2}{2} - 27\lambda_{S_2}^{(1)} \right) \\
&- 9{g_2}^2 \lambda_{S_2}^{(1)} \\
&+ \frac{243{g_1}^4}{4} \\
&+ 4{\lambda_{12}^{(1)}}^2 \\
&+ \frac{{\lambda_{12}^{(A)}}^2}{4} \\
&+ 4{\lambda_{HS_2}^{(1)}}^2 \\
&+ \frac{{\lambda_{HS_2}^{(A)}}^2}{4} \\
&+ 12{\lambda_{S_2}^{(1)}}^2
\end{aligned}
$$

$\lambda_{HS_1}^{(1)}$

$$\begin{aligned}16\pi^2 \frac{d\lambda_{HS_1}^{(1)}}{d\ln\mu} = &\lambda_{HS_1}^{(1)}(6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big))\\ &+ 6\lambda_5\,(\lambda_5)^*\\ &+ 24\mathrm{lambdaT3GeneralQ022000I1}\,(\mathrm{lambdaT3GeneralQ022000I1})^*\\ &+ \frac{9{g_2}^4}{4}\\ &- 9{g_2}^2\lambda_{HS_1}^{(1)}\\ &+ \frac{3{g_1}^4}{4}\\ &- 3{g_1}^2\lambda_{HS_1}^{(1)}\\ &+ 4\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}\\ &+ 6\lambda_1\lambda_{HS_1}^{(1)}\\ &+ 4{\lambda_{HS_1}^{(1)}}^2\\ &+ 6\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}\\ &+ \frac{3{\lambda_{HS_1}^{(A)}}^2}{4}\end{aligned}$$

$\lambda_{HS_2}^{(1)}$

$$\begin{aligned}16\pi^2 \frac{d\lambda_{HS_2}^{(1)}}{d\ln\mu} = &\lambda_{HS_2}^{(1)}(6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big))\\ &+ 6\lambda_5\,(\lambda_5)^*\\ &+ \frac{9{g_2}^4}{4}\\ &- 9{g_2}^2\lambda_{HS_2}^{(1)}\\ &+ \frac{27{g_1}^4}{4}\\ &- 15{g_1}^2\lambda_{HS_2}^{(1)}\\ &+ 4\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}\\ &+ 6\lambda_1\lambda_{HS_2}^{(1)}\\ &+ 4{\lambda_{HS_2}^{(1)}}^2\\ &+ 6\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}\\ &+ \frac{3{\lambda_{HS_2}^{(A)}}^2}{4}\end{aligned}$$

$\lambda_{12}^{(1)}$

$$\begin{aligned}16\pi^2 \frac{d\lambda_{12}^{(1)}}{d\ln\mu} = &3\lambda_5\,(\lambda_5)^*\\ &+ \frac{9{g_2}^4}{4}\\ &- 9{g_2}^2\lambda_{12}^{(1)}\end{aligned}$$

$$
\begin{aligned}
&+ \frac{27{g_1}^4}{4} \\
&- 15{g_1}^2\lambda_{12}^{(1)} \\
&+ 4{\lambda_{12}^{(1)}}^2 \\
&+ 6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)} \\
&+ 6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)} \\
&+ \frac{3{\lambda_{12}^{(A)}}^2}{4} \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}
\end{aligned}
$$

λ_5

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_5}{d\ln\mu} = \lambda_5 &\left(6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big) + \frac{\lambda_{12}^{(A)}}{6} + \frac{2\lambda_{HS_1}^{(A)}}{3} + 2\lambda_{HS_2}^{(A)}\right) \\
&- 9{g_2}^2\lambda_5 \\
&- 9{g_1}^2\lambda_5 \\
&+ 2\lambda_{12}^{(1)}\lambda_5 \\
&+ 2\lambda_1\lambda_5 \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_5 \\
&+ 4\lambda_{HS_2}^{(1)}\lambda_5
\end{aligned}
$$

$\lambda_{HS_1}^{(A)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{HS_1}^{(A)}}{d\ln\mu} = \lambda_{HS_1}^{(A)}&(6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)) \\
&+ \frac{16}{3}\lambda_5\,(\lambda_5)^* \\
&+ 64\mathrm{lambdaT3GeneralQ022000I1}\,(\mathrm{lambdaT3GeneralQ022000I1})^* \\
&+ {g_1}^2\left(6{g_2}^2 - 3\lambda_{HS_1}^{(A)}\right) \\
&- 9{g_2}^2\lambda_{HS_1}^{(A)} \\
&+ \lambda_{12}^{(A)}\lambda_{HS_2}^{(A)} \\
&+ 2\lambda_1\lambda_{HS_1}^{(A)} \\
&+ 8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)} \\
&+ 2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}
\end{aligned}
$$

$\lambda_{HS_2}^{(A)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{HS_2}^{(A)}}{d\ln\mu} = \lambda_{HS_2}^{(A)}&(6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)) \\
&+ 16\lambda_5\,(\lambda_5)^* \\
&+ {g_1}^2\left(18{g_2}^2 - 15\lambda_{HS_2}^{(A)}\right) \\
&- 9{g_2}^2\lambda_{HS_2}^{(A)}
\end{aligned}
$$

$$
\begin{aligned}
&+ \lambda_{12}^{(A)} \lambda_{HS_1}^{(A)} \\
&+ 2\lambda_1 \lambda_{HS_2}^{(A)} \\
&+ 8\lambda_{HS_2}^{(1)} \lambda_{HS_2}^{(A)} \\
&+ 2\lambda_{HS_2}^{(A)} \lambda_{S_2}^{(1)}
\end{aligned}
$$

$\lambda_{12}^{(A)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{12}^{(A)}}{d\ln\mu} = &\frac{4}{3}\lambda_5 \left(\lambda_5\right)^* \\
&+ {g_1}^2 \left(18{g_2}^2 - 15\lambda_{12}^{(A)}\right) \\
&- 9{g_2}^2 \lambda_{12}^{(A)} \\
&+ 8\lambda_{12}^{(1)} \lambda_{12}^{(A)} \\
&+ 2\lambda_{12}^{(A)} \lambda_{S_1}^{(1)} \\
&+ 2\lambda_{12}^{(A)} \lambda_{S_2}^{(1)} \\
&+ \lambda_{HS_1}^{(A)} \lambda_{HS_2}^{(A)}
\end{aligned}
$$

$lambdaT3GeneralQ022000I1$

$$
\begin{aligned}
16\pi^2 \frac{dlambdaT3GeneralQ022000I1}{d\ln\mu} = &\text{lambdaT3GeneralQ022000I1}(6\,\mathrm{Tr}\left(Y_d\ \left(Y_d\right)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ \left(Y_e\right)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ \left(Y_u\right)^\dagger\right)) \\
&- 9{g_2}^2 \text{lambdaT3GeneralQ022000I1} \\
&- 3{g_1}^2 \text{lambdaT3GeneralQ022000I1} \\
&+ 2\lambda_1 \text{lambdaT3GeneralQ022000I1} \\
&+ 8\lambda_{HS_1}^{(1)} \text{lambdaT3GeneralQ022000I1} \\
&+ 4\lambda_{HS_1}^{(A)} \text{lambdaT3GeneralQ022000I1} \\
&+ 2\lambda_{S_1}^{(1)} \text{lambdaT3GeneralQ022000I1}
\end{aligned}
$$